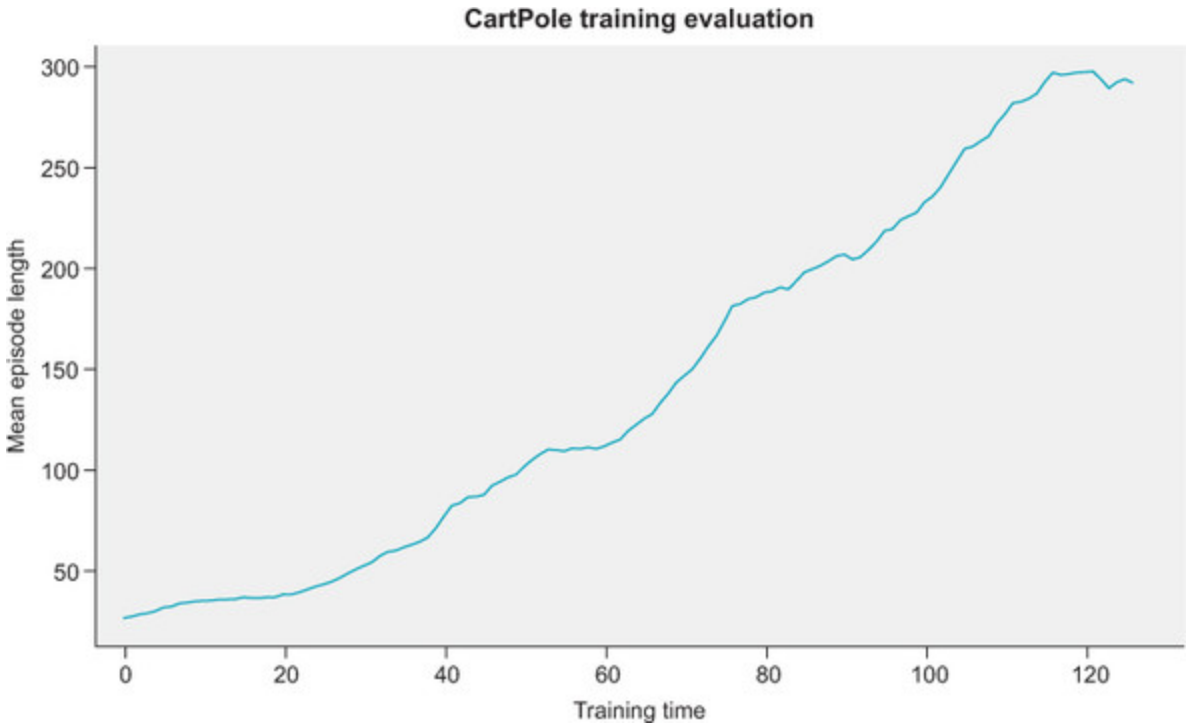


Notice in figure 5.15 that the *N*-step model starts getting better right away and reaches an episode length of 300 (after just 45 seconds), compared to only about 140 for the Monte Carlo version. Also notice that this plot is much smoother than the Monte Carlo one. Bootstrapping reduces the variance in the critic and allows it to learn much more rapidly than Monte Carlo.

As a concrete example, imagine the case where you get 3-step rewards of [1,1,–1] for episode 1 and then [1,1,1] for episode 2. The overall return for episode 1 is 0.01 (with $\gamma = 0.99$) and 1.99 for episode 2; that's two orders of magnitude difference in return just based on the random outcome of the episode early in training. That's a lot of variance. Compare that to the same case except with (simulated) bootstrapping, so that the return for each of those episodes also includes the bootstrapped predicted return. With a bootstrapped return prediction of 1.0 for both (assuming the states are similar), the calculated returns are 0.99 and 2.97, which are much closer than without bootstrapping. You can reproduce this example with the following code.

Listing 5.10. Returns with and without bootstrapping